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A Controlled Comparison of Lysergic Acid Diethylamide (LSD) and Dextroamphetamine in Alcoholics

BY LEO E. HOLLISTER, M.D., JACK SHELTON, M.D., AND GEORGE KRIEGER, M.D.

Seventy-two alcoholic patients were admitted to a controlled comparison of LSD and dextroamphetamine as treatments. In the context of little associated psychotherapeutic intervention, LSD produced slightly better results early, but after six months the results were alike for both treatment groups. Controlled studies of such treatments are not only possible but mandatory, the authors conclude, if one is not to be misled into ascribing special therapeutic attributes to a specific treatment.

THE TREATMENT of alcoholics, by whatever means, has been generally discouraging. Although a small fraction of alcoholics remit, it is difficult to attribute this remission to any particular treatment. Following the old injunction of William James that the "cure for dipsomania is religiomania," many investigators have turned to psychotomimetic drugs to provide the latter. The results reported thus far have been almost unbelievably encouraging.

Alcoholics were treated with single doses of LSD (200 to 400 μg.) or mescaline (500 mg.) and a prolonged interview. Complete abstinence or minimal drinking was obtained in six and no change in 12 patients(5). Single doses of 200 μg. of LSD were given to 68 chronic alcoholics, 26 of whom attained sobriety during an average follow-up period of 38 weeks. Personality factors were explored during the drug-induced state(4). Doses of 400 to 1500 μg. were used along with psychotherapy for 61 alcoholics followed for three to 18 months. Thirty were much improved, while 16 showed some degree of improvement(3).

An attempt to replicate the work of the Saskatchewan group, which has reported the most outstanding results, using a situation as nearly similar as possible, revealed no increase in sobriety following an LSD therapeutic intervention even if as many as three massive doses were given(6). Almost all studies can be criticized on the usual grounds; absence of controls or random assignment to comparison treatments, failure to use blind techniques, failure to account for nonspecific factors in treatment programs, and inadequate follow-up procedures.

While it may be impossible to do a perfectly controlled study of LSD in alcoholics, we felt that some of the objections to previous studies could be overcome. Accordingly, we attempted to study the effects of a single large dose of LSD, compared with a similar dose of dextroamphetamine using blind controls.

Method

Patients

Men between the ages of 21 and 50 years of age who qualified for admission to VA hospitals were selected for study. Prime requirements for selection were these: a) an acute alcoholic episode within the past two weeks; b) the patient must have resided continuously in the area for the past year; and c) the patient must have held

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The drugs used in this study were furnished by Mr. Harry Althouse, Sandoz Pharmaceuticals, Inc. Statistical analyses were provided by the Veterans Administration Western Research Support Center, Edward F. Gocka, Ph.D., director. Dr. Joseph Grismer performed the interviews on the Drinking Behavior Scale.

the same job, either his last or current one, for at least six months. The latter two requirements were primarily for assuring some degree of stability for future follow-up efforts.

These conditions were considered as disqualifying: a) past history of schizophrenic reaction or severe affective disorder; b) present evidence of CNS disorder complicating alcoholism, trauma, or other cause; and c) presence of serious medical illnesses, such as severe hypertension, myocardial infarction, decompensated cirrhosis, or other hazardous condition.

Treatment

It was expected that most patients would require some initial treatment directed at detoxification, requiring parenteral fluids, vitamins, and either chlorpromazine or chloral hydrate or both drugs. As soon as possible after appropriate medical treatment was completed, and no later than one week after admission, the test treatment was given on a "one-shot" basis. No phenothiazine tranquilizer had been administered within the preceding 48 hours.

A minimum period of 24 hours elapsed following the test treatment prior to discharge of the patient, which was accomplished as expeditiously as possible. During the entire period in the hospital, the patient was not assigned to any type of psychotherapy; he participated in ward activity programs as indicated, but the main emphasis on treatment was medical.

The major "psychotherapeutic" intervention prior to administration of the drugs was a discussion with the patient of his problem drinking. The discussion was designed to minimize guilt, pointing out that it was not necessarily some psychological weakness on the part of the patient which led to his problem but rather that he had been hooked by an addicting drug. The focus was not on the solution of specific personality problems during the drug experience but rather on using it as a generally introspective experience.

Within the bounds of medical ethics, patients were given as little concrete information as possible about the drugs to be tested. General instructions were provided in writing as well as orally, but the actual drugs were not named nor any mention made of a control. The two alternate treatments were: LSD-25, 600 μg.; or dextroamphetamine, 60 mg. These medications were administered orally in a similar form, being assigned in a random fashion. While it was expected that patients might ultimately become aware that two treatments were being used and that most certainly staff members would become aware of differences in patients treated, every effort was made to see that patients were treated exactly alike throughout the period.

On the day of treatment, the patient had a breakfast of clear liquids, was taken to a special room on the ward, and was given the assigned treatment at 8:30 a.m. He remained in the room for the next eight hours under constant attendance. Lunch was provided on a tray.

At 4:30 p.m., the patient was taken to his regular bedroom or resumed whatever ward activities he felt up to, although he could not leave the ward. Although insomnia was anticipated, no chloral hydrate was ordered merely for sleep; drugs were to be used in the post-test period only if behavior became disturbed. The patient was encouraged to use this stimulated state for additional self-examination.

The room in which the treatments were given was used only for that purpose. Although on a hospital ward, it looked somewhat "homier" than the usual private room, since it was furnished with a couch, two occasional chairs, and two tables. A source of music was provided and opaque shades allowed darkening of the room.

A research assistant was in constant attendance to provide nonverbal support. That is, no attempt at "psychotherapy" was made; he merely answered direct questions and provided necessary and brief reassurance. The psychiatrist saw the patient for a one-hour period about 30 minutes after the drug was administered. The main purpose of this visit was also for reassurance, emphasizing the previous instruction that the patient was to use the drug experience for self-examination; again, no psychotherapy was given. The patient was seen again for 15 minutes during the fifth hour by the physician, once again to reinforce the introspective goals of therapy.

Evaluations

We felt that the arbitrary use of "abstinence" or "drinking" criteria, so customary in the past, might not be appropriate to most addictions and that some objective measurement of the degree of drinking and its impairment of the patient's social functioning might allow us to detect changes short of these quite global ones. Accordingly, we set about developing a Drinking Behavior Scale, based on the three general areas of drinking habits, social behavior, and occupational adjustment. The items were selected on the basis of discussion among a number of colleagues and deliberately kept to the smallest possible number.

Initially, an arbitrary system of scoring was applied, based on the presumed weight of each item in defining an alcoholic. Later the scale was applied to known alcoholics for verification of the items and the scoring key. Finally, scoring was adjusted to discriminate the most between treatment groups. Despite these further adjustments, most of the clinically selected items and the scoring weights assigned to them were found to be valid and few changes were made. The details of the development of this scale and its contents are reported in another paper(1).

Treatment was evaluated on the basis of an interview with the patient by a psychiatric resident and the subsequent codification of the data on the Drinking Behavior Scale. The entry interview was based on behavior during the previous six months. A subsequent interview was obtained at two months following hospitalization; a second follow-up interview was obtained after six months, again based on the six-month period. Follow-up interviews were scheduled for the next six-month period.

The interviewer was completely divorced from the treatment program and had no knowledge of the treatment received by any individual patient. It was possible that patients might provide clues in the follow-up interviews, but every attempt was made to assure an impartial and unbiased coding of the interview data.

The Drinking Behavior Scale (DBS) was designed to provide a total score indicative of the total degree of impairment due to drinking; this score was achieved by summing the individual items. In addition it was possible to make comparisons on an item-by-item basis as well. Results to be presented are based on comparisons between the two treatments at the two-month and six-month interviews. Although some patients were successfully followed for 12 months, the rate of attrition in the sample at this late date was such that the sample was too selected to be considered as typical of the initial sample.

All of the results are based on the DBS scored with the original scoring key. The mean initial DBS score for the 72 alcoholic patients who entered the study was 74, with a standard deviation of 29 and a range of 20 to 174. By any reckoning, all were problem drinkers, varying from the mildest to the most severe. Only 45 patients were followed for six months, 27 dropping out in that interval.

TABLE 1
Background Variables Among Patients Followed Up and Dropouts

VARIABLE	STUDIED CASES		DROPOUTS	
	LSD (N=26)	AMPHETAMINE (N=19)	LSD (N=11)	AMPHETAMINE (N=16)
Age: median	45	45	46	43
range	31-51	33-51	34-51	34-49
Family history of alcoholism	10	8	7	5
Unmarried	4	4	4	6
Education: elementary	3	3	3	6
secondary	19	12	7	7
college	3	3	1	3
graduate	1	1	0	0
Unemployed	9	10	5	5
No religious affiliation	1	2	1	4

In terms of background variables, there was little difference between the two treatment groups in either the studied cases or dropouts. The latter, however, differed from the former in being more often unmarried and having lower educational attainments. (table 1).

Results

At the two-month follow-up, patients treated with LSD were doing significantly better than those treated with dextroamphetamine (table 2). This result was evident when patients were arbitrarily classified as abstinent (a zero DBS score), social (a score of 10 or less), problem (any score indicating problem drinking), or worse (a score somewhat higher than the initial score). The mean change scores on the DBS also differed significantly; they were 58 for the LSD-treated group and 33 for the dextroamphetamine-treated group ($F = 8.5$; 1 and 50 df; $p < .01$).

Of the original 72 patients treated, 20 were not available for two-month ratings, ten treated with each drug. The initial DBS scores on these dropouts did not differ significantly (mean score for LSD patients, 80.5; for dextroamphetamine patients, 82.2). Consequently, the difference in results from the two treatments at the two-month follow-up was not explainable by some selective attrition of more severely ill patients in one or the other group.

By the time of the six-month follow-up, whatever differences between the treatments that were present at two months had vanished (table 3). This result was evident in the distribution of patients rated as abstinent, social, same, or worse, as well as by the strikingly close mean change scores on the DBS. The change from LSD was 48 and for dextroamphetamine, 49, a clearly nonsignificant difference.

TABLE 2
Two-Month Follow-Up Results in 52 Alcoholic Patients

DBS SCORE CATEGORY	LSD GROUP * (N=29)	AMPHETAMINE GROUP * * (N=23)
Abstinent	8	6
Social	10	5
Problem	10	10
Worse	1	2

* Mean change score on the DBS=58.
* * Mean change score on the DBS=33. ($p < .01$, analysis of variance.)

TABLE 3
Six-Month Follow-Up Results in 45 Alcoholic Patients

DBS SCORE CATEGORY	LSD GROUP (N=26) *	AMPHETAMINE GROUP (N=19) * *
Abstinent	8	7
Social	5	2
Same	12	9
Worse	1	1

* Mean change score on the DBS=48.
* * Mean change score on the DBS=49. (Not significant, analysis of variance.)

As might have been expected, the number of dropouts increased at the six-month period: ten treated with LSD and 17 treated with dextroamphetamine were no longer available for rating. At this time it appeared that the greater number of dropouts in the amphetamine-treated group were patients somewhat less sick initially than the dropouts among the LSD group (initial mean DBS score of 70 as compared with 82).

The results obtained at six months for individual patients are shown graphically in figure 1. It is immediately apparent that while many patients still remained problem drinkers, the degree of their impairment had markedly dropped. Only two of the 45 patients followed for this period of time were worse, all the others showing some degree of improvement.

The number of patients available for follow-up at one year was too small and

Figure 1

Distribution of Changes on the DBS Among 26 Patients Treated with LSD and 19 Patients Treated with Dextroamphetamine at the Six-Month Follow-Up

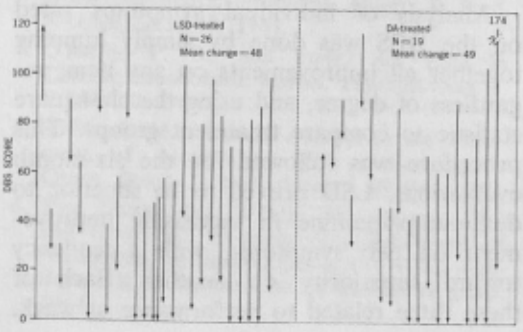


TABLE 4
Changes in Patients Relative to Duration of Follow-Up

DBS CHANGE*	LSD GROUP	AMPHETAMINE GROUP
Patients followed at two months and also at six months		
Worse	7	3
Same	16	10
Better	2	3
Patients followed at six months and also at one year		
Worse	1	0
Same	12	5
Better	4	7

* Worse or better determined by a ± shift of 20 percent of previous DBS score or absolute shift of more than 15 points.

too selected to consider in detail. Only 17 patients treated with LSD and 12 treated with dextroamphetamine were followed for this period. Although it is often customary to use dropouts as a criterion of a better response, this procedure is at best questionable and certainly not applicable for the present study. As might have been expected, the longer term follow-up group was heavily weighted toward those who had made good responses, but if one compares them with the state of patients followed for two months, it also appears that the same phenomenon holds (table 4).

As compared with patients followed for six months, those observed at two months were somewhat better, with more tendency toward relapse at the six-month period; on the other hand, the patients followed for one year tended to be those who showed continued improvement. Thus, the choice of a six-month follow-up period appeared justified as the least biasing and most feasible evaluation period.

Analysis of individual symptoms rated on the DBS was done by simply lumping together all improvements on any item, regardless of degree, and using the chi-square statistic to compare treatment groups. This procedure was followed for the six-month evaluations. LSD proved to be superior to dextroamphetamine in increasing improvement on two symptoms, with a tendency toward superiority on another. Each of these three related to performance at work. Among those not able to work at their expected level of capacity when first seen, 12 of 14 on LSD improved while only five of 11 on dextroamphetamine improved. Among those missing work when first seen, 18 of 22 on LSD improved while only six of 14 on dextroamphetamine improved. Both degrees of improvement were at the .05 level of statistical significance. Among those who had feelings of job inadequacy when first seen, 11 of 16 on LSD improved while only four of 11 on dextroamphetamine improved, a difference at the .10 probability level.

Five of the original 72 patients died within a year of entering the study, all but one, whose cause of death was unknown, from some complications of alcohol (accidents, suicide, or decompensated cirrhosis). Such high mortality rates are common in long-term observations of alcoholics. None of the patients who could be followed reported additional use of psychotomimetic drugs.

In the great majority of cases, the deemphasis on verbal interaction and the encouragement of "letting go" led to rather surprisingly tranquil drug experiences. Occasionally, moderate agitation required additional attention from the psychiatrist and a temporary reduction of sensory input, as by encouraging a supine position and reducing light and music. Only two patients were sufficiently agitated to require intramuscular administration of 50 mg. of chlorpromazine. During one of these periods of agitation, a grand mal seizure occurred, without injury, in a patient with a previous history of "rum fits." Vomiting occurred in two other cases, but even nausea was surprisingly infrequent.

Because of the publicity given the so-called "prolonged LSD reaction" it should be emphasized that all but three of these patients were discharged within 48 hours following their drug experience and after a careful mental status evaluation. For two of the three who stayed longer, hospitalization was briefly extended in order to allow time to arrange housing and other social considerations. Moderate confusion was present in the third patient, requiring hospitalization for some further 48 hours. A late recurrence of psychotomimetic drug ef-

fects was not mentioned by any patient at the follow-up interviews.

Discussion

It should be emphasized that these results can only be interpreted in relation to the conditions of the experiment. This study was primarily designed to test the original contention that LSD given to alcoholics would produce a profound and moving experience per se, and that with little or no specific psychotherapy a favorable response could be obtained.

Since then, a number of variants to the use of LSD have been added, emphasizing greatly the therapeutic preparation of the patient prior to the drug experience and with much therapeutic intervention during it(2). These new variants in technique may indeed be more rational, but from the data we have obtained in this study, it would not appear that LSD as a primary treatment offers any advantage over dextroamphetamine.

The most striking aspect of the present study was the fact that both treatment groups improved to a rather considerable degree, which, in the absence of any control or comparison group, would have been construed as a remarkable effect of the single treatment being evaluated. To some extent, we had anticipated improvement: at the time of the initial interview these patients were close to some trouble caused by their alcoholism and could scarcely do anything else except improve at some later date. Another possible explanation for the improvement is that the difficulties in follow-up with such patients may lead to selective dropping out among those who show poorer results. No doubt some of this occurred, but we have no evidence that it was a biasing factor for or against the treatments being compared.

A third explanation might be one based on the phenomenon of early remorse and late denial, quite commonly seen in alcoholics. When in trouble, alcoholics often exaggerate their foibles and, when out of it, minimize them. Still another explanation could be that both drug treatments were specifically effective. The dose of dextroamphetamine was large enough to produce quite appreciable effects, which, although only slightly mimicking those of LSD, might still have been therapeutic. This explanation seems unlikely, however.

Finally, it might be that our milieu is specifically effective in treating alcoholics. The general approach to alcoholics here has been the alleviation of guilt (in rather sharp contrast to the approach of Alcoholics Anonymous) and it could be that this approach may have specific benefit. At the moment this explanation cannot be supported or denied but, in view of the many attitudinal approaches to alcoholics which have been tried in the past and have failed, one would not offer such an explanation with much assurance.

It seems clear that any assessment of treatments for alcoholism demands some kind of control or comparison group if one is not to be misled. In the past, controls have not been considered feasible. In the present study we feel that the controls were adequate even though not perfect. Other situations may demand more ingenuity in devising controls, but few should present such formidable problems that none are possible.

REFERENCES

1. Hollister, L. E., Shelton, J., and Gocka, E. M.: The Drinking Behavior Scale: An Attempt to Quantify Alcoholic Impairment, unpublished.
2. Kurland, A. R., Unger, S., Shaffer, J. W., and Savage, C.: Psychedelic Therapy Utilizing LSD in the Treatment of the Alcoholic Patient: A Preliminary Report, Amer. J. Psychiat. 123:1202-1209, 1967.
3. MacLean, J. R., MacDonald, D. C., Byrne, U. P., and Hubbard, A. M.: The Use of LSD-25 in the Treatment of Alcoholism and Other Psychiatric Problems, Quart. J. Stud. Alcohol 22:34-45, 1961.
4. O'Reilly, P. O., and Funk, A.: LSD in Chronic Alcoholism, Canad. Psychiat. Ass. J. 9:258-260, 1964.
5. Smith, C. M.: A New Adjunct to the Treatment of Alcoholism: The Hallucinogenic Drugs, Quart. J. Stud. Alcohol 19:406, 1958.
6. Van Dusen, W., Wilson, W., Miners, W., and Hook, H.: Lysergic Acid Treatment of Alcoholism, prepublication report #325, California State Department of Mental Hygiene, 1964.